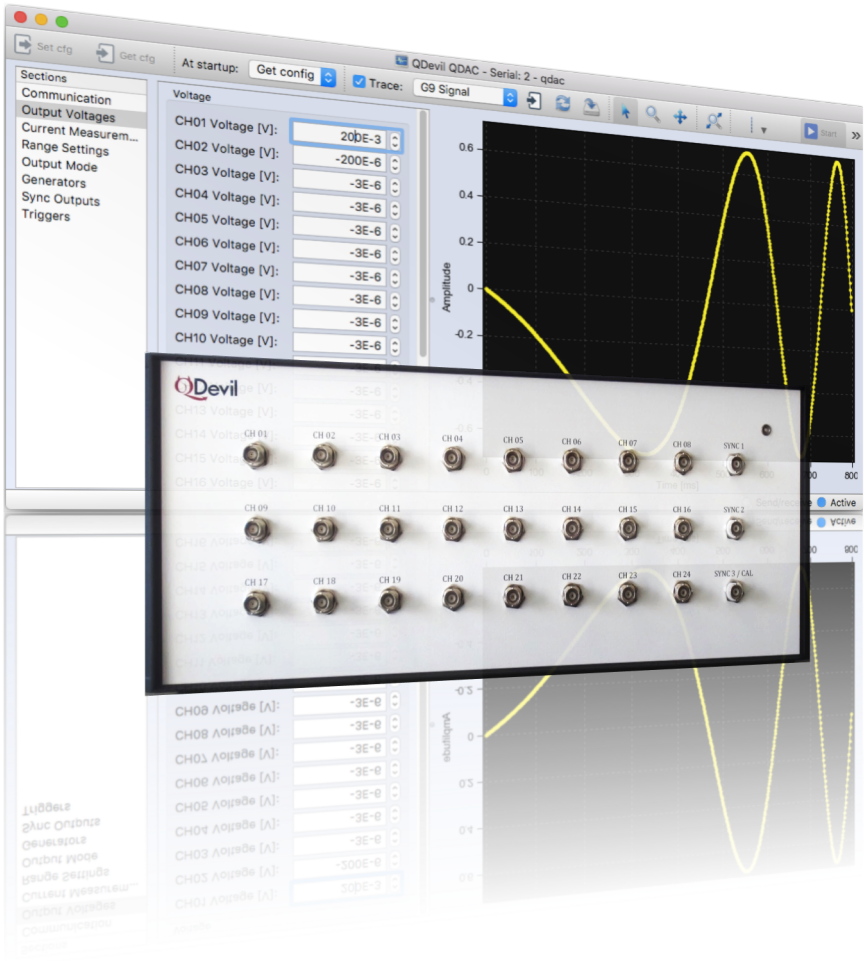
QDAC Labber Driver



Author: AK
Date: 09-05-2019

Manual version: 1.0b
Driver version: 1.01

1 Installing the driver

The driver consists of the following files which are zipped together:

QDevil QDAC.ini	The Labber driver definition file (default driver)
QDevil QDAC With SW Ramping.ini	Alternative Labber driver definition file enabling software sweeping*
QDevil_QDAC.py	The custom Python code part of the driver
qdac.py	Python interface to the QDAC, version 1.22
AWGcurve.txt	Example curve for the AWG

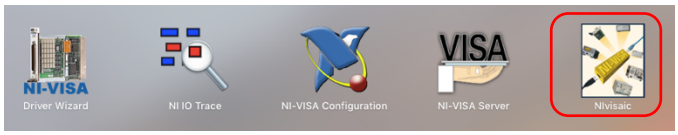
These files should be placed in a subfolder “QDevil QDAC” in the Labber driver directory in your user folder or in the driver folder in the Labber installation folder. Please see the Labber documentation for details. The driver has been tested with Labber version 1.6.3

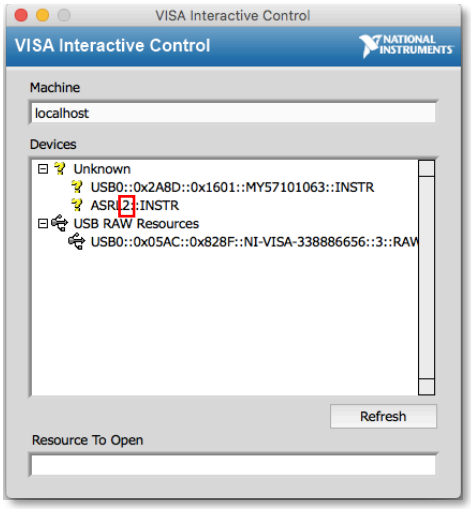
*)In order to minimize the number of controls in the driver user interface “QDevil QDAC.ini” driver does not enable software ramping. If software ramping between values in step sequences is desired, please use the “QDevil QDAC ramp.ini” driver instead – otherwise you may delete it.

2 NI-VISA – finding the device address

The QDAC Labber driver uses the NI VISA framework for communication using usb/serial port on the computer. So, it is necessary to install NI VISA on the computer running Labber. Please see the Labber documentation for details.

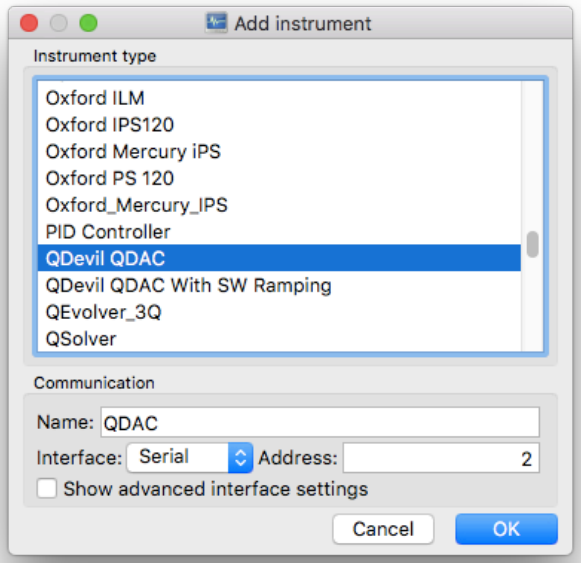
To find the port number of the QDAC, please run “NIvisaic” (NI VISA Interactive Control). The QDAC will be listed as “ASRL##::INSTR”, where ## is the port number. In case you have multiple usb/serial port devices, you may need to unplug and re-plug their communication wires to find out which is which.





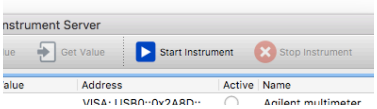
3 Add the QDAC to the Instrument Server

In the Labber Instrument Server, add the QDAC by clicking the “Add” button. The driver is named “QDevil QDAC” (the alternative driver is “QDevil QDAC With SW Ramping”). Select it in the list, enter a name, and enter the serial port number found in NI VISA in the Address field. Finally click OK.



4 Starting the QDAC

Adding the QDAC to the Labber Instrument Server allows browsing all the available options and functionality of the driver and generating a configuration file; but, it does not initiate actual communication with the instrument. Push the “Start” button in the Instrument Server dialog or in the Config dialog to connect to the QDAC.



“Starting” the QDAC takes about 15 seconds for a 24 channel QDAC and 30 seconds for a 48 channel QDAC, because the Labber driver will first read the complete state of the QDAC – except for the curve data for the AWG as this is not currently readable. Especially reading the current sensors takes time. You can see when Labber has completed reading the QDAC state by watching the LED on QDAC front panel: When its stops blinking red, the communication is completed.

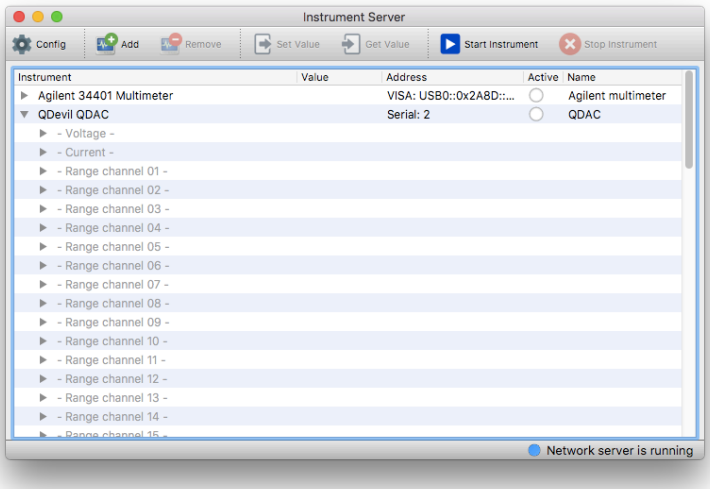
The reason for reading in the current state of the QDAC is to be able to recover from computer- or software crashes.

If you do not want the driver to read-in the hardware state at start-up, edit the QDevil QDAC.ini file and delete or comment out the “startup: Get config” line in the top section. Then remove the QDAC from the Instrument Server and add it again in order to apply the changes to the ini file.

5 Controlling the QDAC

The described version of the Labber driver provides access to most of the QDAC's. Functionality, not accessible is for example instrument calibration and soft synchronization. Please see the QDAC manual for a full description of the QDAC's functionality.

The QDAC can, as other Labber instruments, be controlled directly from the Instrument Server dialog using the "Get Value" and "Set Value" commands for each quantity. All QDAC quantities (settings, output voltages, and measured currents) are listed in various expandable groups in the Instrument Server dialog. All output voltages (24 or 48) are placed in one group and all measured currents in another. This makes it easy to overview them both in the Instrument Server dialog and in the Measurement dialog than if all options for each channel were placed together.



5.1 The Config dialog

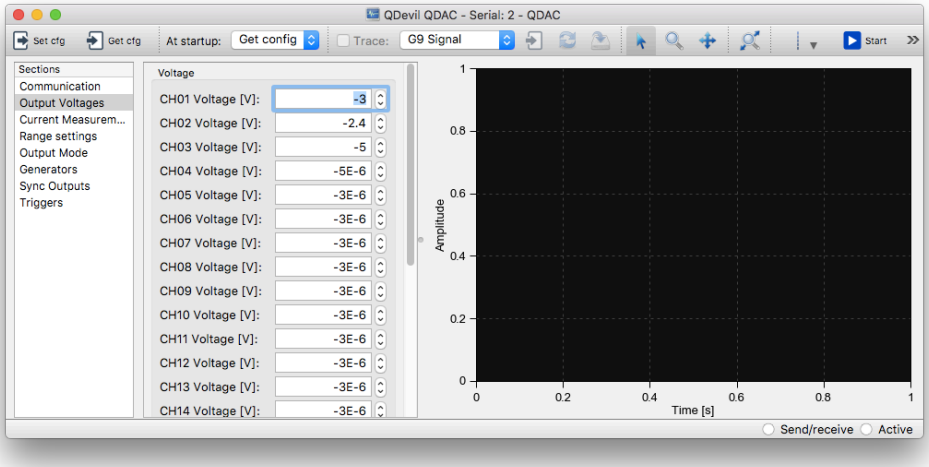
The Config dialog provides a convenient front panel for the QDAC. In here the settings and values have been grouped further into sections, making it easier to find controls for this instrument which has a much larger number of input/outputs than most other instruments have. Please consult the QDAC manual for functionality details.

In the left side of the Config dialog you find the sections into which the various controls are grouped. When one of these is clicked the corresponding controls appear in the middle.

In the right side there is a chart area. This is used for displaying any curve uploaded to the QDAC's arbitrary waveform generator and appears empty until a curve has been loaded from disk.

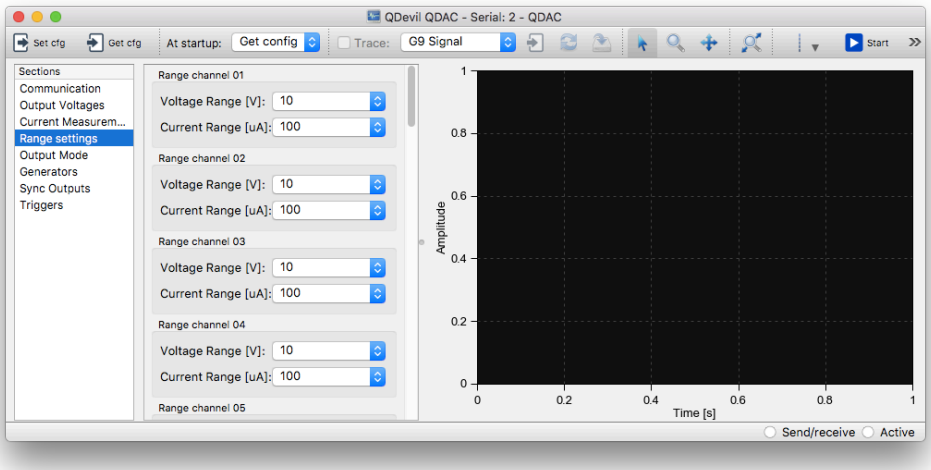
Example: Setting a voltage

To set a voltage on one of the QDAC outputs, simply go to the “Output Voltages” section and enter the new voltage in the field for the desired channel.



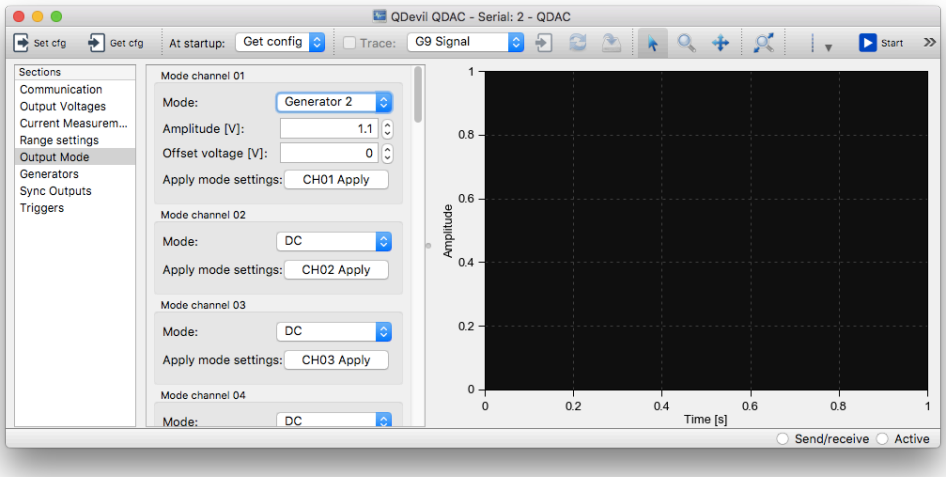
To change the output voltage range (or the current measurement range) go to the “Range settings” section.

Note that the QDAC cannot be in the low voltage range and high current range at the same time.



Example: Outputting a waveform

To output a waveform, first go to the “Output mode” section. In the Mode pull down menu select which generator should be connected to this channel. Then set the amplitude and offset for the generator (see QDAC manual). Finally push the “CH## Apply” button to transfer the “Mode” settings to the QDAC. The reason why the Mode setting is not applied immediately when it is changed is the following: IF a generator is already running then switching a channel’s Mode to output that generator could result in applying an undesired high amplitude or offset to the output. So therefore, one needs to push “Apply”, as this gives you a chance to first adjust offset and amplitude.

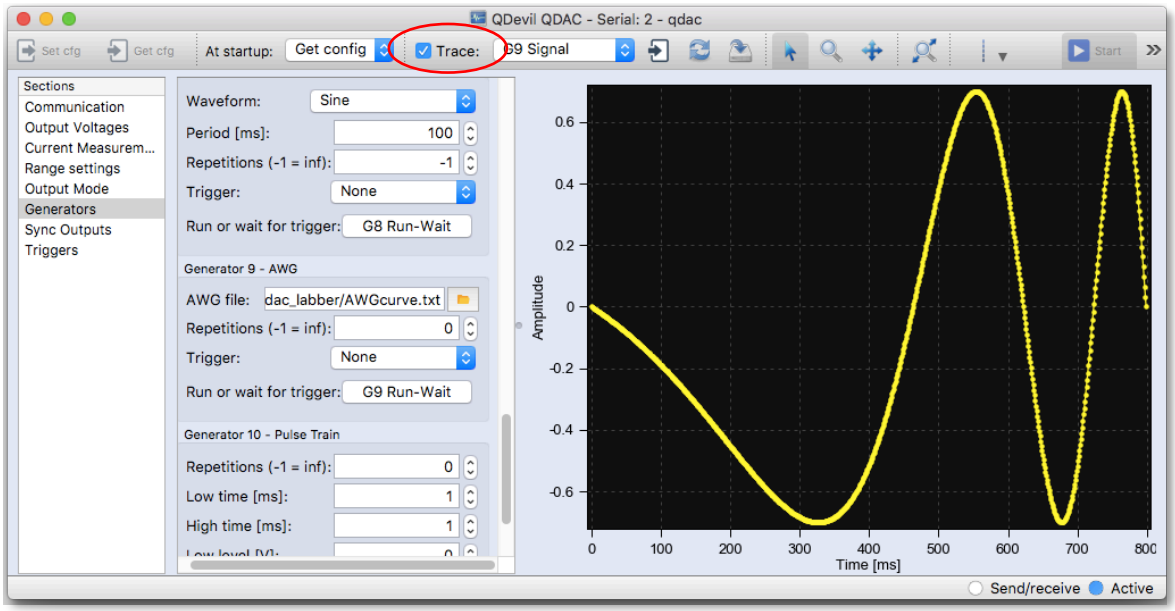


Then got to the “Generators” section and configure it. To program and start the generator using the displayed leave the “Trigger” field to “None” and push the “Run-Wait” button. If you want to start several generators at the simultaneously select the same trigger for those generators. Then got to the “Triggers” section and push the relevant trigger “Fire” button.

Note: Setting repetitions to zero and pushing the “Run-Wait” button will stop a running generator. If “Trigger” is different from “None” the generator will also stop when the “Run-Wait” button is pushed.

Uploading a curve data to the AWG generator

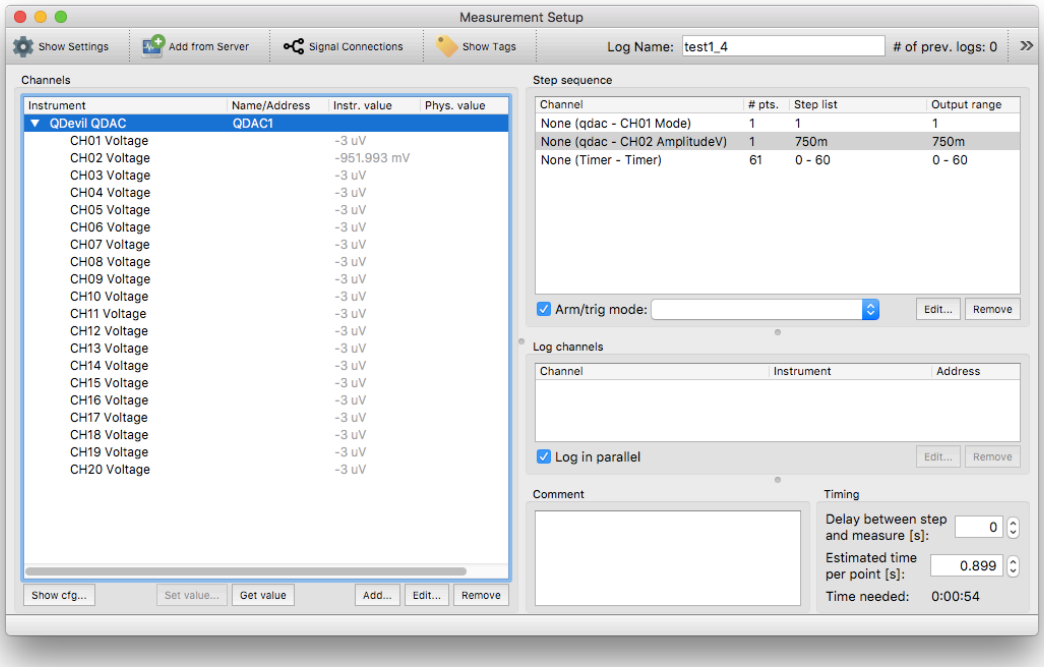
To upload an array of curve data to the arbitrary waveform generator, click the “AWG file” icon and browse to a file which has one column of numerical values defining the samples for the AWG. Select and open the file. The check the “Trace” check box in the menu bar in the dialog. This will actually load and display the curve. Each sample will be 1 ms (one sample clock) long .

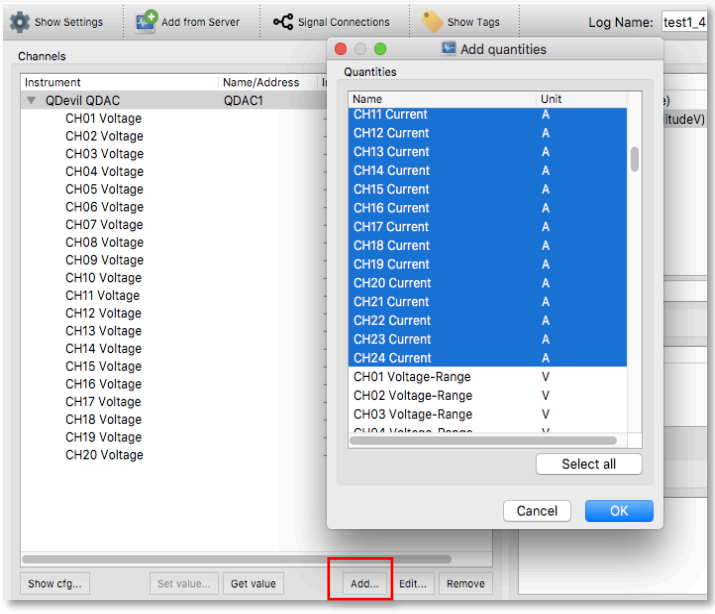


Now set the number of repetitions, and optionally a trigger, and click the “Run-Wait” button. This will upload the curve to the QDAC and start the AWG – or arm it if a trigger was chosen.

Measurement dialog – adding more channels

In Labber version 1.6.3 and earlier the Measurement dialog will only show up to 20 quantities per instrument as default. This means that initially you will only see the voltage setting of channels 1-20. To enable all channels and also for example the current sensors, select the QDAC in the Measurement dialog, and click the “Add” button and add more channels.

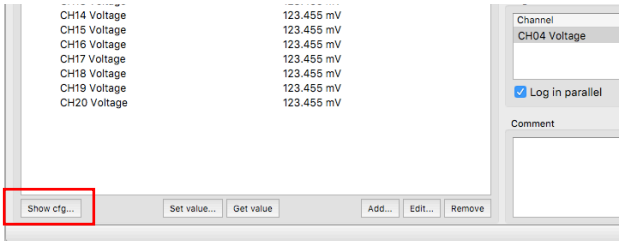


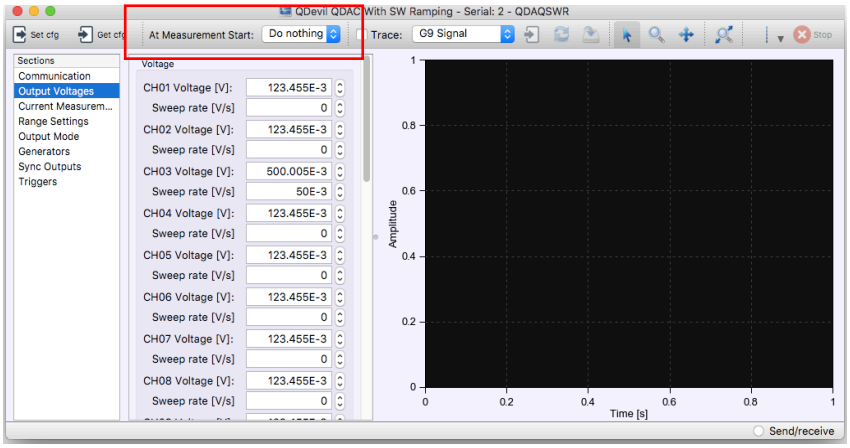


Note that push buttons cannot be added to a step sequence. Therefore, function generators cannot be programmed and started, and the output mode of channels cannot be changed within a step sequence.

When “Starting” a measurement the instrument will initialize. For the QDAC it means that its current state will be read. This can take some time. To avoid go to the Config dialog by pushing “Show cfg…” in the Measurement dialog, and then set the “Set Measurement Start” option to “Do nothing”. This is not recommended for general usage. But when testing sequences, it may become annoying having to wait for

the Labber driver to read the QDAC state. Alternatively, the “Get config” behavior may be disabled entirely by changing the “startup” option in the ini file as mentioned earlier.





The Config dialog when started from the Measurement dialog. Note that in this screen shot there are sweep rates listed for each output voltage. This is because the “QDevil QDAC With SW Ramping.ini” driver was chosen instead of the default driver.